

Modjaw in daily orthodontic practice: about a case treated with FAS aligners

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Resumen

En las últimas décadas, y gracias a los avances tecnológicos, los métodos de registro de dinámica mandibular han sufrido un constante desarrollo, pasando de ser realizados de un modo analógico a ser, hoy en día, llevados a cabo con una secuencia completamente digital. En esta publicación se hace un repaso de la evolución que ha tenido este campo en los últimos años, haciendo énfasis en la última tecnología digital 4D Modjaw. Se detallan las claves del registro de la dinámica mandibular y los conceptos fundamentales a tener en cuenta para conseguir unos óptimos resultados clínicos. Por último, se presenta un caso clínico ilustrado con Modjaw donde se ha empleado una férula de desprogramación tipo FACE a fin de resolver la sintomatología articular de la paciente y un tratamiento definitivo mediante ortodoncia con alineadores FAS.

Introduction

Imagine that we never used our teeth. Do you think we would have tooth wear, gingival recession, abfractions, muscle problems or signs and symptoms of joint problems? The answer, without a doubt, is "NO". Whether you have a dental class of I, II or III is not important. If we did not use our teeth to chew, we would never have the abovementioned problems. As Dr. Congost from Germany says, *"It's all about chewing"*¹. It is only when we chew that the teeth begin to undergo occlusal forces. That is why good occlusal function (orthopedic stability) should be the number one goal with any dental treatment. If we want to achieve a correct chewing pattern, avoid tooth wear, prevent recessions and abfractions, as well as avoid TMJ problems, orthopedic stability is of vital importance^{2,3,4}. Function is synonymous with orthopedic stability. As Dr. Jeff Okeson⁵ says: "Orthopedic stability exists when the intercuspal position of the teeth fits with the musculoskeletal position of the condyles in the glenoid fossae." Therefore, the first step is to determine the patient's stable condylar position. In this regard, dynamic recordings greatly benefit diagnosis. Axiography provides a dynamic recording of condylar trajectories, and makes it possible to determine the patient's axis of rotation. The location of the stable condylar position, in which a condylar rotation occurs with a single arc of closure, is

Abstract

In recent decades, and thanks to technological advances, mandibular dynamics recording methods have undergone constant development, going from being carried out in an analogical way to being carried out today with a completely digital sequence. This publication reviews the evolution that has taken place in this field in recent years, emphasizing the latest 4D Modjaw digital technology. The keys to the mandibular dynamics registry and the fundamental concepts to be taken into account to achieve optimal clinical results are detailed. Finally, a clinical case illustrated with Modjaw is presented where a FACE-type deprogramming splint has been used in order to resolve the patient's joint symptoms and a definitive treatment by orthodontics with FAS aligners.

extremely important for diagnosing and planning any orthodontic or rehabilitation treatment. It is on the patient's arc of closure, and in an orthopedically stable position, where predictability is achieved in movements where the objective is mandibular autorotation.

Historical evolution

Recording the condylar axis of rotation and mandibular dynamics has been studied for decades. Its beginnings date back to the beginning of the 20th century. By 1905, George Campion was already emphasizing the importance of locating and using the mandibular axis of rotation as a reference for treatments. This author devised a system for recording the location of the condylar axis of rotation to study condylar trajectories (Fig. 1A)⁶. In the years that followed, various authors such as Bennett, Gysi, and McCollum developed works and designed devices to locate the hinge axis and study mandibular dynamics^{7,8,9,10,11,12} (Fig. 1B). Various analog systems appeared throughout the 20th century, in response to the need to work using the patient's axis of condylar rotation and true arc of closure. Some of the most popular were the Panadent (Fig. 1C) and Cadiax systems; a first attempt to digitize the method but still based on analog procedures (Fig. 1D). These devices used a facebow connected to the mandible and markers on both sides that described the mandibular movement on graph paper affixed to the patient's skull, in the case of Panadent; for Cadiax, a

digital program was used^{13,14}.

The recording process began by placing a bite fork in the patient's mouth and attaching a facebow to establish the axio-orbital axis. The graph was affixed to each of the hinges, on which the axis of rotation and the condylar trajectories were determined.

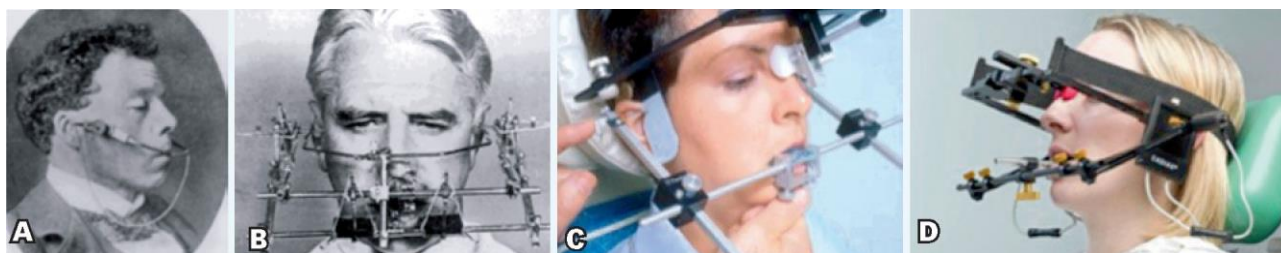
Different dynamic recordings were made to determine the location of the axis of rotation, its route being marked on the graph.

Once the recordings were made, this information was transferred to a facebow and later to an articulator mount.

This is how recordings of the axis of rotation, lateral translation movement, and Bennett movement were achieved, to individualize the articulator.

The analog axiography process, although it reproduced

the axis of rotation and mandibular dynamics, was subject to minor errors at the different stages of recording, meaning that the final result was significantly altered. These errors, which are undetectable during axiography, completely changed the patient's diagnosis and treatment plan. This limitation in terms of accuracy and reproducibility also had other disadvantages: longer time in the chair, a complex learning curve, increased patient discomfort, and complex laboratory procedures. These factors led to mixed results that discouraged wide-scale adoption by the profession. While necessary, this information has generally been overlooked in clinical practice, although in light of the information obtained using the new digital methods, it is a key factor in the quality of long-term results.



Figs. 1 Historical evolution. A) Graphic facebow (Campion, 1905); B) Gnathograph (McCollum and Stuart, 1937); C) Panadent system; D) Cadiac system

Digital axiography: MODJAW

The latest technological advances have led to the development of new methods for recording mandibular dynamics in a digital format, offering greater ease for mounting and recording, greater predictability and reproducibility, and less time in the chair. The equipment has also become more comfortable for both professional and patient, along with the advantages of working with a fully digital sequence.

Modjaw is a system for recording and studying mandibular dynamics in order to diagnose and plan orthodontic or rehabilitative treatment¹⁵.

It uses a digital camera to reproduce true mandibular movements in the arches in the .STL files, which is useful for assessing condylar trajectories, improving diagnostic accuracy, and designing oral appliances that are perfectly customized from a functional aspect. This device has been made possible by developing two sectors of technological research: one recording movements using a sophisticated 3D stereophotogrammetry-based camera, the other displaying digital models of the dental arches, which can also now be paired with the patient's cone beam

computed tomography (CBCT) (Modjaw version 4.0), integrating a kinematic assessment and display of the temporomandibular joints (TMJ), thus increasing the accuracy and predictability of the result in complex cases¹⁶.

The device is based on recordings of mandibular movements via IR sensors, which comprise the following (Figs. 2): a computer with an infrared camera; jaw position sensor (Tiara); mandibular position and movement sensor (Butterfly); bite fork; and marker (Tally).

Position and movement sensors affixed to the patient are detected by the computer's infrared camera. This software reproduces the position and movement information in real time. The sensor on the head is useful for the position in the maxilla and the axio-orbital axis space, while the mandibular sensor marks the position of mandibular movements and records them.

It is important to record different types of movement cycles: opening and closing, centric relation, lateral, protrusive, speech, mastication, and border movements. To obtain a correct diagnosis, the same movement must be repeated several times, thus detecting if there is instability in a condylar path.



Figs. 2. Modjaw 4D system. 2A-B) Modjaw computer; C) Infrared camera; D) Maxillary position sensor (Tiara); E) Marker (Tally); F) Bite fork; G) Mandibular position sensor (Butterfly)

Displaying the mandibular dynamics

The condylar trajectory can be analyzed from different planes to display the patient's recordings. The graphs show the plane of view via an icon: sagittal, axial, or frontal.

The range traveled is displayed simply using the grid graph, which is calibrated in 1 mm spaces. This tool provides us with a very practical visual measurement of

the movements recorded.

The study of mandibular dynamics can be very precise. We can dynamically see the movement of rotation or translation that the condyle makes, and how much movement. We can also compare the movement of both condyles simultaneously and the trajectory of the incisal point (Fig. 3).

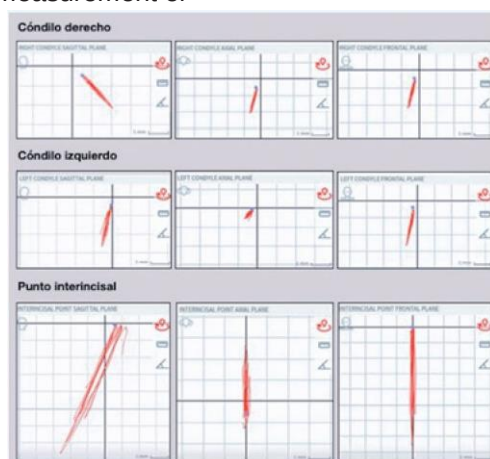


Fig. 3. Graph plotting of the condylar trajectories and interincisal point in the sagittal, axial, and coronal planes.

Cóndilo derecho	Right condyle
Cóndilo izquierdo	Left condyle
Punto interincisal	Interincisal point

Occlusal contacts

Recording mandibular dynamics makes it possible to study the occlusal contacts in each of the movements made by the patient, following a map of collisions either in monochrome, or colored to reflect distance.

The monochrome study highlights a situation similar to contacts that would be marked in the mouth with articulating paper, indicating only the areas where there is occlusal contact (Fig. 4).

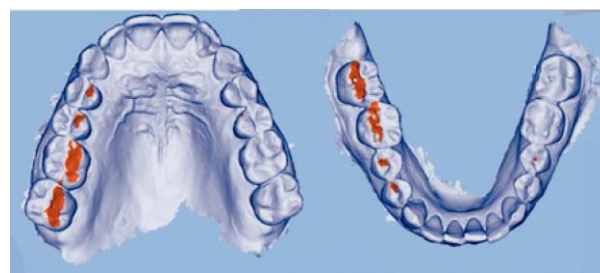


Fig. 4. Dynamic display of occlusal contacts and inter-arch distances. Monochrome map. It is useful for displaying just the contact areas, showing a similarity with the intraoral verification of the occlusal contacts marked with articulating paper.

The color map is useful for precisely calibrating occlusal contacts and inter-arch distances. Detects from 0.5 mm of disocclusion in blue to red, where there is contact. Each color change is equivalent to 0.1 mm closer proximity (Fig. 5).

The occlusal contacts can also be studied statically, establishing a static virtual relationship of the models. In this way, we can detect any premature contacts, for example, in the patient's arc of closure.

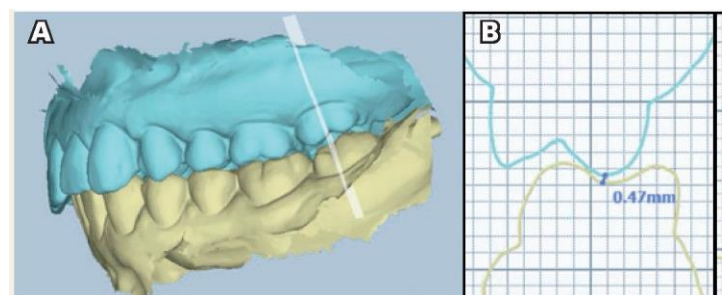
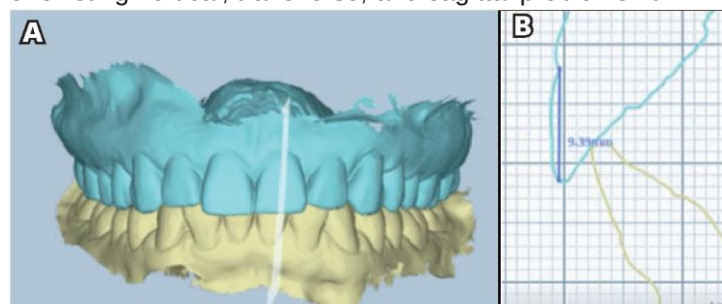
Hinge axis

One of the key factors in axiographic recording is establishing the arc of closure. Using points located on the patient's skin, corresponding to both condyles, the center of rotation is indicated arbitrarily. These points determine the horizontal axis of the arc of closure (Fig. 6).

When recording the condylar trajectory, it is essential to determine the moment at which the condyle is performing a stable rotation movement. If the points have not been precisely positioned on the patient's skin, the arbitrary axis can be corrected because the Modjaw software detects if there is rotational movement by recalculating the hinge axis points (Fig. 7).

Plane of section

To best analyze the occlusion in the desired relationship, the upper and lower casts can be studied by planes of section. Sectional planes provide a clearer assessment of existing vertical, transverse, and sagittal problems. It



Mandibular movements

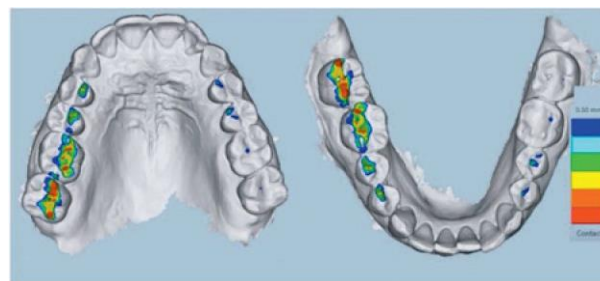


Fig. 5. Dynamic display of occlusal contacts and inter-arch distances. Color map for distances. In red: areas of contact with the antagonist. The rest of the colors indicate a greater or lesser proximity to the antagonist tooth.

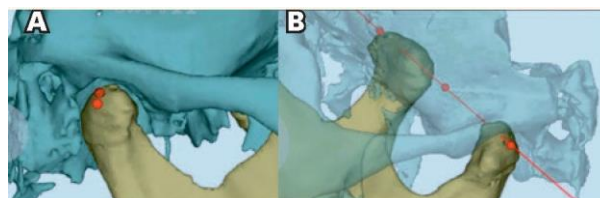


Fig. 6. Condylar axis of rotation. A) Arbitrary axis; B) Hinge axis

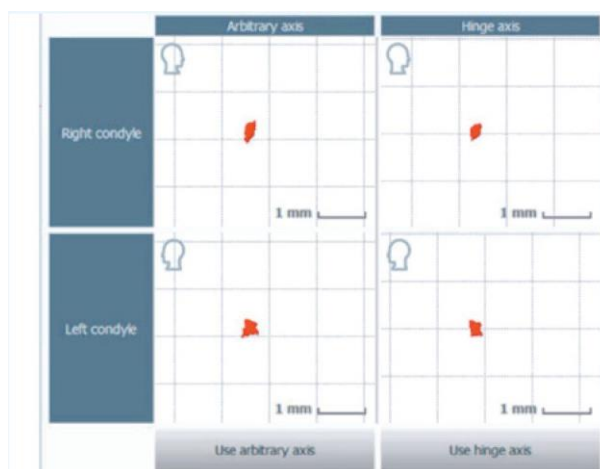


Fig. 7. Hinge axis detection

is also possible to change the orientation of the slices and perform longitudinal and angular measurements on the models in each slice (Figs. 8 A-B and Figs. 9 A-B).

Figs. 8. Plane of section in the anterior sector. A) Display on the model. B) Display of the sectional plane with length and angulation references.

Figs. 9. Plane of section in the posterior sector. A) Display on the model B) Display of the sectional plane with length and angulation references.

Rotating movement

Condylar rotation movement can be assessed when recording the mandibular arc of closure.

The condylar position is stabilized using the Dawson maneuver to record the mandibular arc of closure (Fig. 10).

If the patient has a correct condylar rotation, we will see how, in the graphs corresponding to the condyles, the blue point does not exceed one of the gridded graphs, which is calibrated in 1 mm spaces.

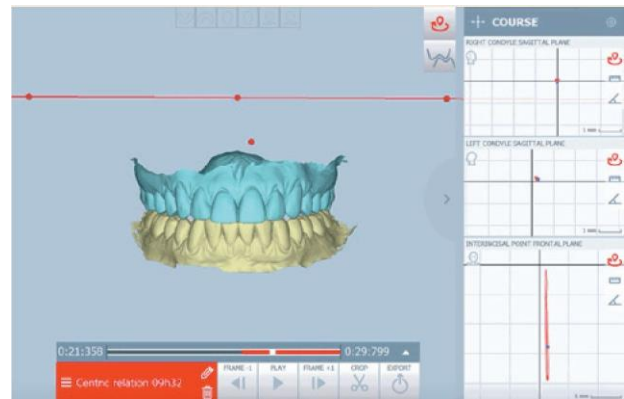


Fig. 10. Condylar rotating movement.

Translation movement

After a certain degree of mouth opening, approximately 10 mm in most cases, the condyle begins to move on the articular eminence. The translational movement

describes a forward and downward trajectory in the sagittal plane, which in the axial and frontal plane is also angled from lateral to medial (Fig. 11).

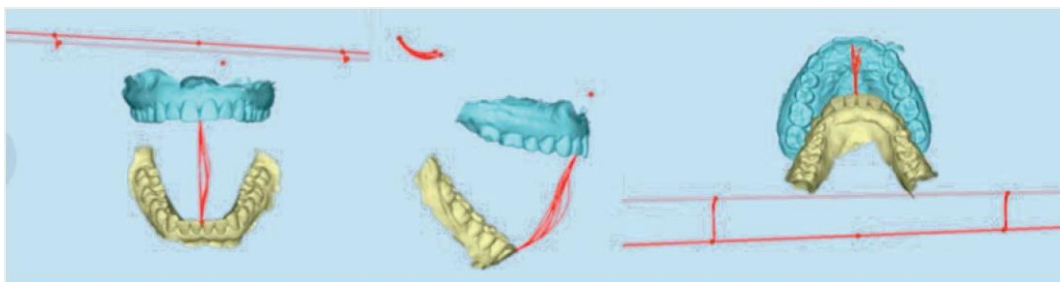


Fig. 11. Translation in a maximum opening movement. Note the different routes in both condyles and in the interincisal point, created by each mandibular movement.

When the translation movement describes the same trajectory during each opening/closing, it is an indication of stability.

Protrusion movement

The mandibular protrusion movement, involving central incisor contact, generates a similar trajectory for both condyles, which slide, guided by the glenoid cavities and teeth. The condyle descends, following the articular eminence (condylar guidance). The inclination of the eminence will influence the condyle's degree of vertical displacement.

The angle at which the condyle descends along the articular eminence during eccentric movements determines the height of the cusps of the posterior teeth. Modjaw calculates the actual condylar tilt of the patient from the protrusive and lateral recording. These data can be used for programming the articulator either in digital or analog.

To calculate the angulation produced by this path, Modjaw performs an automatic calculation, taking the most anterior and inferior part of the translational movement as the vertex (Fig. 12).

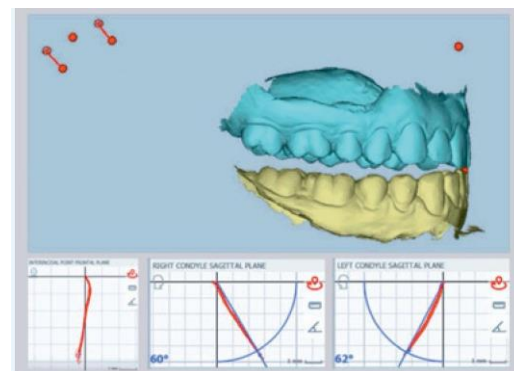
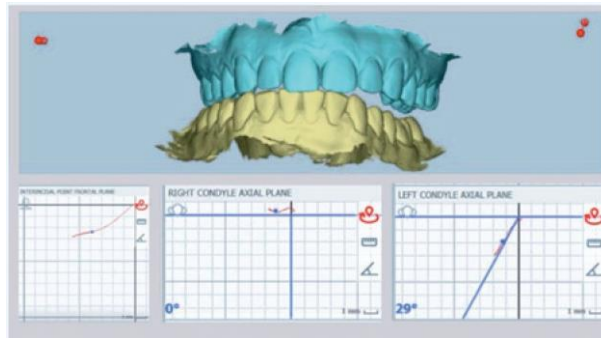


Fig. 12. Condylar guidance obtained from the protrusive recording. Edge-to-edge dental position (anterior end of movement). Graph (frontal view) of the protrusive path. Graph (sagittal view) of the condylar guidance. Angulation calculated by Modjaw.

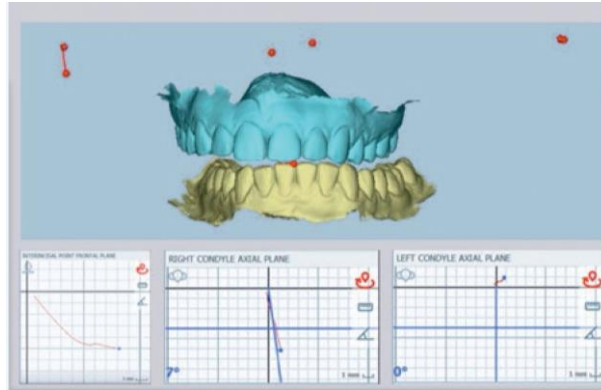
Lateral movement

If the movement is lateral, the condyle moves downwards and medially on the balancing side (Bennett angle), while the condyle on the working side rotates. During lateral movement, due to the laxity of the articular ligaments, a lateral translation is produced towards the same side of the movement, potentially generating interference between the posterior cusps, with a

significant effect on their anatomy. The immediate Bennett side shift is measured at the condyles as the distance from the maximum occlusion position to the working occlusion position. This distance varies between 0.5 and 2.5 mm, the average being 1.5 mm. Modjaw records the Bennett angle of the balancing side condyle from the corresponding lateral recording (Figs. 13 and 14).



Figs. 13. A). Right laterotrusion recording. B) Lateral Bennett movement in the right condyle. C) Bennett angle in the left condyle.



Figs. 14. A) Left laterotrusion recording. B) Bennett angle in the right condyle. C) Lateral Bennett movement in the left condyle.

Clinical case

A 15-year-old female patient comes to our office suffering from temporomandibular dysfunction. On the first visit we verified the existence of an unstable occlusion.

After performing diagnostic recordings, the patient was diagnosed with skeletal class I with a tendency to class III, good facial profile, slight maxillary hypoplasia, slight deviation of the mandible to the right, reverse smile curve, posterior gummy smile and lack of exposure of incisors at rest (Figs. 15).



Figs. 15. Pre-treatment extraoral photographs

At the intraoral level, the patient presented with molar and canine class I, upper and lower gaps, lower midline deviated to the right, lack of overjet and overbite, and good arch shape (Figs. 16).



Figs. 16. Pre-treatment intraoral photographs.

In the lateral telerradiography of the cranium and cephalometric tracing, we can observe a class I skeletal pattern with a tendency to III and excessive torque of the upper and lower incisors. (Fig. 17)

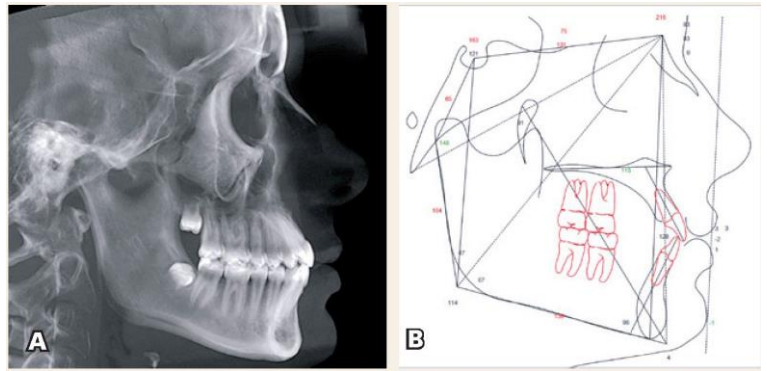


Fig. 17. A) Lateral telerradiography of the cranium before treatment. B) CEPH: cephalometric tracing.

In the panoramic view we observe good periodontal health, the existence of the four wisdom teeth and a suspicion of condyle malposition outside the glenoid fossa (Fig. 18).

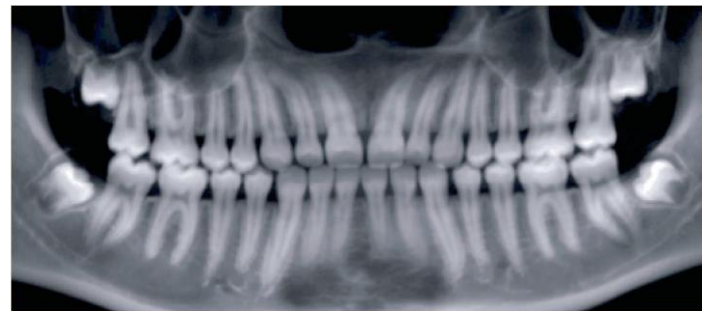
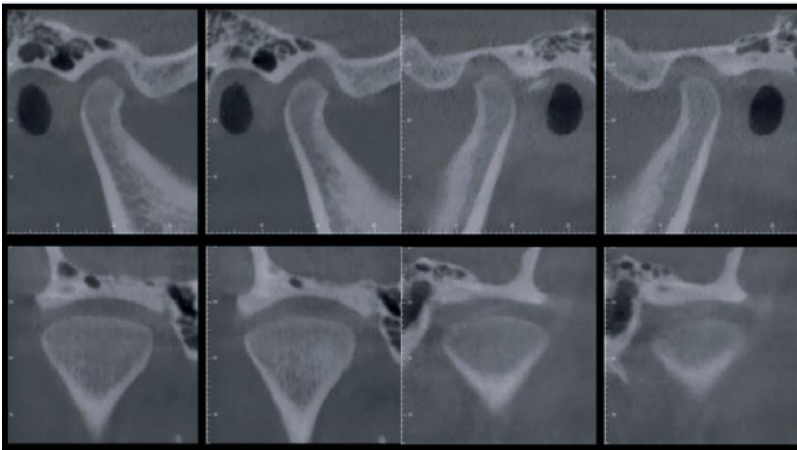


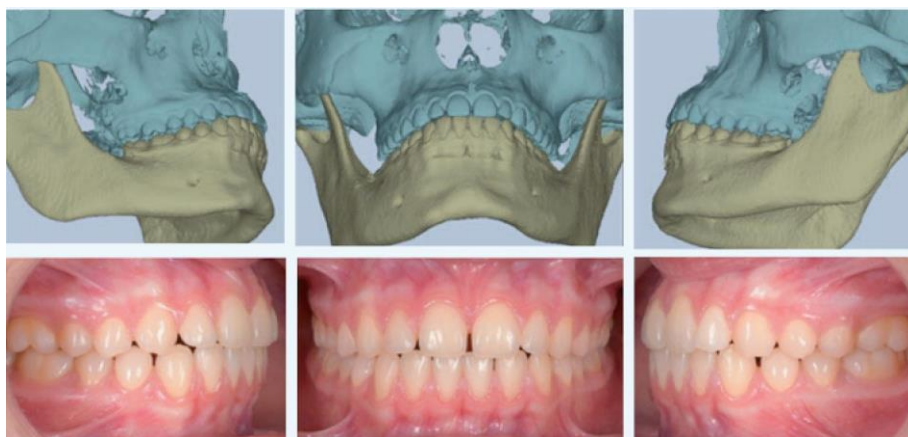
Fig. 18: Panoramic x-ray before treatment.

This suspicion was confirmed with the sagittal and coronal slices of the TMJ in the CBCT, where we see how the condyles are correctly corticalized but descended with respect to the articular eminence. (Figs. 19)



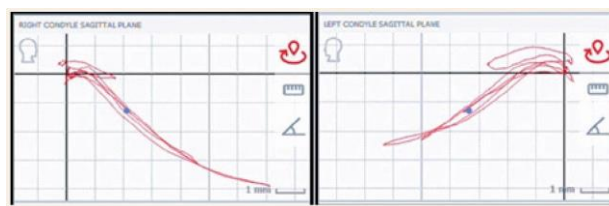
Figs. 19. Sagittal and coronal CBCT slices of the right and left TMJ.

In the virtual model assembly (obtained from Modjaw), we observed a slight discrepancy between the patient's maximum intercuspation and her arc of closure (Figs. 20).



Figs. 20. Discrepancy between the patient's arc of closure (top) and maximum intercuspation (bottom).

In the analysis of the Modjaw graphs during the opening and closing movements, we can observe how, during translation movements, the condyles never follow the same condylar trajectory, an indicator of orthopedic instability (Figs. 21).



Figs. 21. Graphs of condylar movements during opening and closing obtained with the Modjaw 4D device. Note how, during translation movements, the condyles never follow the same condylar trajectory, an indicator of orthopedic instability.

Another indicator of this instability is revealed by the graphs obtained from Modjaw when manipulating the patient in "centric relation", where we observed the inability of her condyles to rotate within the fossae (Figs. 22).



Figs. 22. Graphs of condylar movements during manipulation of the patient's arc of closure (centric relation) with the Modjaw 4D device. Note the inability of her condyles to rotate within the fossae.

Given the issues listed above, we made the decision to begin treatment using a FACE deprogramming splint, in order to stabilize the musculoskeletal position of the joints and create a single mandibular arc of closure (Figs. 23).



Figs. 23. FACE deprogramming splint. A) Rear part: the patient wears it during the day; B) Front part: the patient wears it at night.

After wearing the splint for a period of five months, all her joint signs and symptoms disappeared and we obtained a single arc of closure. Following this first phase of joint treatment, we performed all the diagnostic

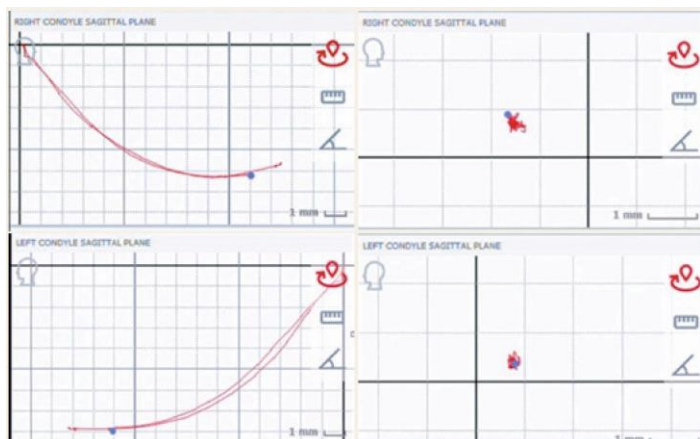
recordings to re-assess the case.

Now, we can see how the occlusion has changed. A greater bilateral class II and greater anterior open bite are observed. (Figs. 24)



Figs. 24. Occlusal relationship of the patient following the splint phase.

The graphs obtained from Modjaw following this first splint phase show a radical change in the patient's condylar movements, both in the "opening-closing" and "centric relation" movements. Now the condyles are able to rotate within the eminence and always translate following the same condylar trajectory. (Figs. 25)



Figs. 25. Graphs of condylar movements during opening and closing (left), and during manipulation of the patient's arc of closure (centric relation) using the Modjaw 4D device (right) following the splinting phase.

The next phase consisted of the orthodontic treatment itself in the patient's arc of closure. The FAS aligners (Forestadent, Pforzheim, Germany) were selected for treatment.

A VTO was performed (Fig. 26) in order to guide our treatment and give correct instructions to the planners, consisting of:

- Alignment and leveling
- Diastema closure

- 2-3 mm posterosuperior intrusion
- Upper incisor positive torque of 5°
- Lower incisor negative torque of 10°

The movement simulation program (OcclusalDesign) of the FAS aligners simulates mandibular anterorotation, taking into account the patient's true hinge axis; this means that we can make our plans for the patient's arc of closure. (Fig. 27)

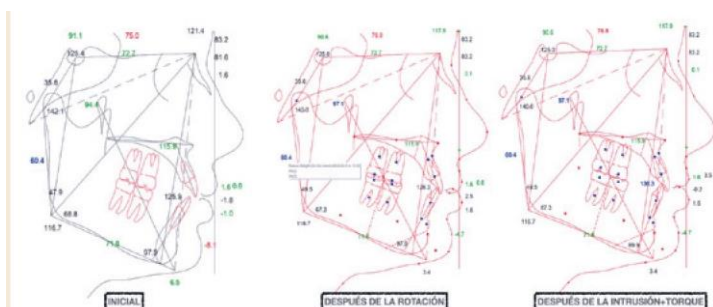


Fig. 26. VTO.

INICIAL	INICIAL
DESPUES DE LA ROTACION	AFTER ROTATION
DESPUES DE LA INTRUSION+TORQUE	AFTER INTRUSION + TORQUE

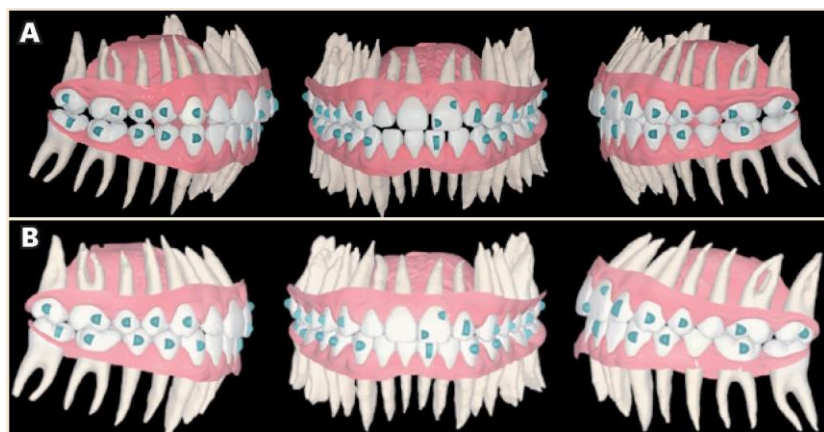


Fig. 27. Final position planned using the FAS aligner movement simulation program (OcclusalDesign).

A) Pre-treatment. B) Post-treatment.

After analyzing the movement table provided by OcclusalDesign, we verified the need to introduce skeletal anchorage in order to be able to create an effective intrusive movement of the upper posterior teeth.

This first phase consisted of 23 upper and lower aligners and lower IPR to achieve retrusion of the lower incisors and centering of the lower midline. Six microscrews were

placed (between the first and second molar buccal, between the first and second premolar buccal, and bilaterally between the second premolar and first molar lingual). The patient was instructed to place 5/16-8 oz elastics between the microscrews and over the aligners to promote the intrusion movement of the maxillary posterior teeth (Figs. 28).



Figs. 28. Intra-treatment intraoral photographs with FAS aligners and 6 microscrews for vertical control.

We can see how with this first phase we have already achieved most of our treatment objectives without distalization or elastics, only with the intrusion of the upper posterior teeth (vertical control) and mandibular autorotation (Figs. 29).



Figs. 29. Resulting occlusal relationship after treatment with the first phase of aligners.

However, to achieve a better finish for the patient, additional aligners were requested in order to improve the molar and canine class on both sides, center the midlines and achieve a greater anterior overbite. The patient followed a second phase of 10 stages, in which she was instructed in the use of intermaxillary elastics (with a class III component on the right side and a class II component on the left side) (Figs. 30).



Figs. 30. Second phase with Class III intermaxillary elastics on the right side and Class II on the left side.

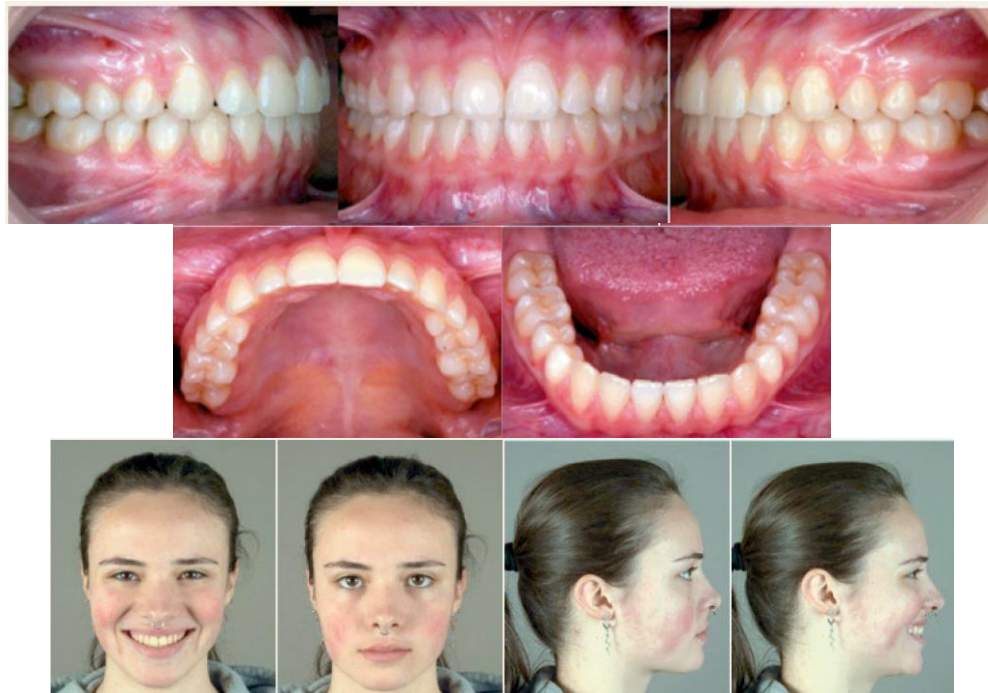
The case ended with the placement of the FAS Finisher, built in the patient's arc of closure to achieve a good finish (Figs. 31).



Figs. 31. FAS Finisher (gnathologic positioner) developed for the patient's arc of closure.

The case was closed with the patient completely asymptomatic, having corrected her occlusal instability and treated the issue with her true arc of closure. In addition, the patient has an improvement in her smile,

and a class I molar and bilateral canine. The midlines are centered and there is good overjet and overbite, and good arch forms (Figs. 32).



Figs. 32. Post-treatment intraoral and extraoral photographs.

The case ended with good root parallelism and no root resorption. The patient will be referred to her dentist for the extraction of the third molars (Fig. 33).

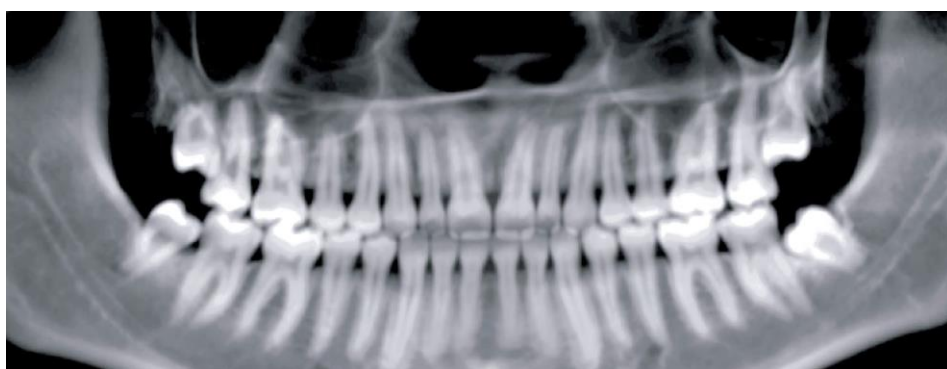


Fig. 33. Post-treatment panoramic x-ray.

In the lateral teleradiography of the cranium and cephalometric tracing we observed a normalization of the torques of the upper and lower incisors, with an improvement in the interincisal angle and a closure of the facial axis thanks to the mandibular autorotation achieved. (Fig. 33)



Fig. 34. Lateral teleradiography of the cranium after treatment.

Discussion

Although traditionally both specific x-rays and articulated models have been the standard in static relationships for the diagnosis of malocclusions and articular pathology, dynamic recordings provide great advantages in temporomandibular joint study.

The study of movement helps us to understand the behavior of the joint and also its pathology. Analog axiographic procedures were subject to errors that were difficult to assess. Modjaw's 4D digital sequence minimizes errors in dynamic recordings and, being more visual, facilitates its understanding, minimizing the learning curve and being user-friendly in daily practice. For the correct diagnosis and treatment of any orthodontic or prosthodontic procedure, it is necessary to consider a stable condylar position and mandibular

dynamics in the arc of closure of the condylar hinge as a starting point and primary objective. We will obtain orthopedic stability by working on the patient's true axis of rotation.

Conclusions

Recording the mandibular dynamics is a key procedure in the correct approach and planning of orthodontic or rehabilitation treatment. The latest technologies facilitate the recording procedure and optimize the accuracy and interpretation of the results. Modjaw is a tool that paves the way for the study of mandibular dynamics and for diagnosis and treatment planning using a digital sequence, firstly taking into account the stability of the patient's condylar position and arc of closure.

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